

# SemOrder: Cost-Based Logical Ordering of Semantic Operators

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CS 59200 - Data Management for AI | Midterm Presentation

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# Motivation: Why Does Operator Order Matter?

## Order A: Title Filter → Plot Filter

Step 1: Run AI on 1,000 short movie titles (~5 tokens each) → 800 pass

Step 2: Run AI on 800 long movie plots (~300 tokens each) → 200 pass

✓ **Cheaper** — short text processed on the larger set

## Order B: Plot Filter → Title Filter

Step 1: Run AI on 1,000 long movie plots (~300 tokens each) → 200 pass

Step 2: Run AI on 200 short movie titles (~5 tokens each) → 160 pass

✗ **Expensive** — long text processed on the full 1,000 rows

**Same result. Very different cost. The order matters enormously.**

- Each filter requires a real LLM (AI) call — these cost real money and time
- A suboptimal ordering on 1,000 rows can waste 5× more tokens and dollars
- Prior systems either ignore this or use expensive trial-and-error to solve it

# Problem Definition

*Given a set of semantic operators  $S = \{s_1, s_2, \dots, s_n\}$  and an input relation  $R$ , find the permutation of  $S$  that minimizes total execution cost (tokens / latency / money) while preserving query correctness.*

- Semantic operators are logically commutative — any order gives the same final result
- But costs are not symmetric — order dramatically affects total tokens consumed
- Cost depends on three unknown factors: input length, selectivity, and operator accuracy

| Factor       | Why It Matters                                                             |
|--------------|----------------------------------------------------------------------------|
| Input length | Longer text = more tokens = higher cost per row                            |
| Selectivity  | A highly selective filter applied first reduces rows for all later filters |
| Accuracy     | An inaccurate filter may appear selective but produces wrong results       |

# Proposed Methodology & Baselines

Baseline 1 — Prior Work

## Local Token-Only

- Zero LLM calls for planning
- Estimates cost using only text length
- Ignores selectivity and accuracy

✗ Gets ordering wrong when selectivity dominates

Baseline 2 — AERO

## Adaptive Sampling

- Runs full LLM calls during execution to learn statistics
- Adapts ordering online as data accumulates

✗ Expensive startup overhead — pays full cost just for planning

Our Contribution

## Bounded Small-Sample Estimation

- Runs LLM on just 10–20 sample rows per operator before execution
- Estimates selectivity + accuracy + token cost from those samples
- Uses all three factors jointly to pick optimal ordering

✓ Fixed planning cost regardless of table size

✓ More informed than Baseline 1, cheaper than Baseline 2

# Evaluation Plan

## Dataset

- IMDB movie database via the Join Order Benchmark (JOB)
- 1,000 movies per configuration
- 25 configurations: language selectivity  $\times$  genre selectivity at {0%, 25%, 50%, 75%, 100%}
- Two text attributes: short title (~5 tokens) and long plot summary (~300 tokens)

## Models

- GPT-4o-mini (remote API): monetary cost and token experiments
- Llama 3.1-8B (local on Chameleon GPU): latency and compute experiments

## Metrics & Comparison

- Total tokens consumed, total latency, total monetary cost
- Ordering decision accuracy vs. oracle (ground truth optimal order)
- Three-way comparison per config: Local-Only vs. Small-Sample (ours) vs. Oracle

## Ablation Study

- Vary sample budget: 5, 10, 20 tuples
- Plot: ordering accuracy vs. planning cost tradeoff curve

# Updated Project Timeline

| Week    | Dates        | Tasks                                                                     |
|---------|--------------|---------------------------------------------------------------------------|
| Week 9  | Mar 10–14    | Midterm presentation, read professor's papers, repo and environment setup |
| Week 10 | Mar 17–21    | Data pipeline, LLM wrappers, reimplement local cost model baseline        |
| Week 11 | Mar 24–28    | Reproduce prior paper results on IMDB/JOB dataset                         |
| Week 12 | Mar 31–Apr 4 | Implement small-sample estimator, integrate selectivity and accuracy      |
| Week 13 | Apr 7–11     | Full experiments across all 25 configurations, generate result figures    |
| Week 14 | Apr 14–18    | Ablation study, tradeoff curve analysis, write-up                         |
| Week 15 | Apr 21–25    | Final report completion in Overleaf, figure polish                        |
| Week 16 | Apr 28       | Submit final report                                                       |

**Key Risk: Infrastructure setup (Chameleon GPU + LLM API wiring) — starting immediately this week.**

# Final Report Draft Status

| Report Section                                           | Status                                       |
|----------------------------------------------------------|----------------------------------------------|
| Introduction                                             | <input checked="" type="radio"/> In Progress |
| Related Work (AERO, Palimpzest, JOB paper, arxiv papers) | <input checked="" type="radio"/> In Progress |
| System Design & Methodology                              | <input type="radio"/> Not Started            |
| Experimental Setup                                       | <input type="radio"/> Not Started            |
| Results & Analysis                                       | <input type="radio"/> Not Started            |
| Conclusion                                               | <input type="radio"/> Not Started            |

*Introduction and Related Work are the current focus — covering the JOB benchmark, AERO, Palimpzest, and the two most recent arxiv papers provided by the professor.*

**Draft available in Overleaf — ready to share during the presentation.**